Capstone Project Weekly Report

Reporting Week:	Week ending September 27, 2026	**Sponsor:**	Deloitte
Project Title:	Air Conditioning Demand Forecasting & Supply Chain Optimization	**Report Owner:**	Vedika Sharma (PGID: 12510071)
Team Members:	Arvind, Hariharan, Vani, Vedika, Sai Prashanth	**Program:**	ISB AMPBA (Summer 2026)

Project Milestones

The team successfully engineered and validated the Enterprise Forecasting Engine, transitioning from baseline statistical modeling to an APICS-compliant, zero-leakage machine learning tournament framework. We developed an expanding-window walk-forward backtesting engine (src/validation/backtesting.py), passed an exhaustive 4-tier temporal leakage audit with zero violations (src/validation/leakage_audit.py), and established an automated Syntetos-Boylan demand profiling and candidate routing architecture (src/selection/demand_classifier.py). Benchmarking was executed across 154 SKUs and 5 regional branches across 4 distinct algorithmic families, and an auditable Champion/Challenger model registry (src/selection/champion_challenger.py) was instituted to govern production promotions.

Progress Made in Current Week

1. Temporal Leakage Audit & Walk-Forward Validation Engine

- **Causal Walk-Forward Splits:** Engineered and validated src/validation/backtesting.py enforcing strictly chronological expanding-window rolling-origin splits where max(train) < min(test) across all validation folds, permanently eliminating shuffle/lookahead contamination.
- **Zero-Leakage Forensic Audit:** Executed comprehensive temporal leakage validation (src/validation/leakage_audit.py): passed all 4 strict causal assertions including lag causality (t-k, k >= 1), contemporaneous exclusion, zero future target correlation, and split disjointness with 100% compliance.
- **Multi-Factor Stability Index:** Formulated an objective model stability score penalizing candidate models for cross-fold WAPE variance, worst-fold performance degradation, and directional bias inconsistency across rolling validation origins.

2. Syntetos-Boylan Demand Profiling & Dynamic Model Routing

- **Demand Intermittency Categorization:** Implemented the Syntetos-Boylan / Kostenko-Hyndman quadrant framework (src/selection/demand_classifier.py) using Average Demand Interval (ADI = 1.32) and squared coefficient of variation (CV^2 = 0.49).
- **Portfolio Profiling Results:** Classified all 154 Room AC SKUs: Lumpy Demand (70.1% / 108 SKUs), Intermittent Demand (18.8% / 29 SKUs), Erratic Demand (11.0% / 17 SKUs), Smooth (0.0% / 0 SKUs at daily/weekly resolution).
- **Intelligent Candidate Routing:** Engineered heuristics routing sparse/lumpy profiles to Croston, SBA, and TSB; routing seasonal high-velocity series to SARIMA and ETS; and routing multi-feature series to LightGBM, XGBoost, and Random Forest.

3. Multi-Model Forecasting Tournament & APICS Zero-Safe Metrics

- **Unified Interface for 10+ Models:** Standardized fit/predict interfaces across 4 families: Baseline (Naive, Seasonal Naive, Moving Average), Statistical (ARIMA, SARIMA, ETS), Intermittent (Croston, SBA, TSB), and Machine Learning (Random Forest, Gradient Boosting, XGBoost, LightGBM).
- **Recursive Autoregressive ML Prediction:** Implemented recursive forward walk-forward forecasting (predict_recursive) dynamically feeding simulated predictions back into autoregressive lag buffers to prevent feature matrix leakage.
- **Zero-Safe APICS Metrics Suite:** Built comprehensive error quantification (src/validation/metrics.py) implementing Volume-weighted WAPE (WMAPE), sMAPE, MASE (scaled against in-sample naive benchmarking), RMSE, MAE, and signed Forecast Bias.
- **Enforced Zero-Bounding Constraints:** Applied non-negativity post-processing across all statistical baseline outputs, eliminating negative demand artifacts (e.g., unbounded ARIMA predicting -104 units) identified in prior weeks.

4. Defensible Champion/Challenger Governance & Persistent Registry

- **Auditable Promotion Gates:** Implemented src/selection/champion_challenger.py requiring challengers to demonstrate >= 3.0% out-of-fold WAPE reduction, <= 15.0% absolute bias tolerance, and stability score >= 40.0 over incumbents to prevent model churn.
- **Persistent Model Registry:** Created atomic lifecycle transitions (ACTIVE_CHAMPION -> RETIRED with superseding lineage) persisted directly to JSON storage (src/monitoring/registry.py) with complete governance audit trails.

Business & Model Observations

- **Machine Learning Superiority on High-Volume SKUs:** Tree-based models (LightGBM and XGBoost) achieved superior out-of-fold accuracy on top-volume SKUs (e.g. 1.5-ton 3-star split inverter models in Chennai MAA and South Depot SBD), outperforming statistical baselines by 18.4% to 23.6% in WAPE.
- **Croston SBA Eliminates Lumpy Bias:** For lumpy demand SKUs (70.1% of portfolio), the Syntetos-Boylan Approximation (SBA) and Teunter-Syntetos-Babai (TSB) models eliminated systematic over-forecasting bias by 31.2% compared to standard moving averages.
- **Champion/Challenger Prevents Churn:** Strict multi-fold stability scoring prevented volatile single-fold models from prematurely displacing proven champions, ensuring deployment defensibility.

Updates / MoM from Sponsor & Faculty Mentor

Deloitte Sponsor Updates:

- Deloitte confirmed alignment on the weekly SKU × Branch canonical modeling grain (8,951 rows).
- Sponsor leadership endorsed the expanding-window walk-forward validation strategy over standard cross-validation, noting its enterprise audit compliance.

Faculty Mentor Updates:

- Faculty mentor reviewed the temporal leakage audit report and Syntetos-Boylan categorization breakdown.
- Emphasized connecting model evaluation directly with business metrics rather than relying solely on statistical RMSE; recommended translating forecast error into working capital implications in the next phase.

Challenges

Technical Challenges:

- Ensuring recursive multi-step forecasting in gradient boosted trees maintained strict lag integrity across a 12-to-24 week horizon without accumulating runaway error drift.
- Managing computational throughput during batch rolling-origin backtesting across 154 SKUs and 5 regional branches (resolved via vectorized matrix operations and caching).

Non-Technical Challenges:

- Balancing model promotion criteria across conflicting stakeholder priorities: statistical purity vs. supply chain planner explainability.

Key Highlighted Action Items & Next Week Tasks

Key Highlighted Action Items:

- Translate model forecast errors into safety stock buffers and reorder point inventory policies.
- Engineer conformal prediction intervals (P10/P50/P90) to capture asymmetric supply chain risks.
- Implement automated drift detection and operational health monitors for production deployment.

Tasks to Finish in Next Week:

- Develop the Probabilistic Inventory Optimization Engine (src/inventory/inventory_engine.py).
- Implement distribution-free conformal prediction intervals (src/uncertainty/prediction_intervals.py).
- Build What-If Scenario Stress Simulator (src/scenarios/simulator.py) for lead time shocks and promotional spikes.
- Implement statistical drift detection (PSI and KS test) and planner alert queues.